

Charles Stephen Thompson

Infrastructure Engineer · Automation & Platform · IAM & SSO

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Infrastructure engineer with 14+ years designing, automating, and operating production platforms — identity, CI/CD, DNS, high-availability databases, monitoring, and incident response — now with LLM agents in the loop. I build the interfaces and guardrails that let automation act on production safely: high-availability PostgreSQL (Patroni) with automated failover, DNSSEC-signed split-horizon DNS, rootless container platforms, and custom MCP (Model Context Protocol) servers exposing git/CI, metrics, DNS, and SIEM to AI agents under least-privilege scopes. Backed by a decade of enterprise IAM/SSO and platform engineering, and a 40+ service private cloud serving dozens of client environments where the automation has real consequences.

EDUCATION

Bachelor of Arts, Modern American Legal History

The University of Chicago — Chicago, IL

2005 - 2010

BA Thesis on the misuse and abuse of copyright law as compared to that seen in the patent field.

SKILLS

AI & AUTOMATION

Claude / OpenAI / ChatGPT / LLM APIs

MCP Serversn8nOpenWebUI

OllamaRAGPrompt Engineering

HOME & BUILDING AUTOMATION

Home AssistantHomebridgeHomeKit

MatterThreadZigbeeControl4

Z-Wave

IDENTITY & ACCESS

IAMSAML / OIDC / OAuth2 (PKCE)

OktaActive DirectoryKerberos/LDAP

ADFSPocket IDGSuite

AUTOMATION & CONFIGURATION

AnsibleChefCloudFormation

JenkinsMDMTerraformCircleCI

GitHub ActionsForgejo Actions

Renovate

DATABASES & HIGH AVAILABILITY

PostgreSQLPostgreSQL HA (Patroni)

MariaDB GaleraReplication & Failover

TimescaleDBInfluxDBHAProxy

PROFESSIONAL EXPERIENCE

Principal Automation & Platform Engineer

BKHD Engineering (platform engineering arm of Brooklyn Helpdesk) — Brooklyn, NY

Jan 2022 - Present

- Architected an AI automation platform where n8n workflows orchestrate LLM agents (Ollama) to triage monitoring alerts, reconcile DNS, and drive CI/CD across dozens of client environments — auto-remediating ~70% of alerts without human intervention and cutting mean time-to-detect by an average of 14 minutes
- Designed, built, and deployed 150+ custom MCP (Model Context Protocol) server integrations exposing Forgejo git/CI, Grafana metrics, Technitium DNS, and a Wazuh SIEM to AI agents under per-scope least-privilege authorization — solving the largely unsolved problem of letting agents act on production safely
- Built and deployed a Go/Vue network log-analytics service end to end — TimescaleDB schema, syslog ingestion at ~100 GB/day, and OIDC/PKCE authentication — replacing a Python predecessor under sustained production load
- Engineered a high-availability Kubernetes/Podman container platform (systemd quadlets) spanning 6 self-hosted and 10 cloud nodes and hosting 40+ client-facing services at a 99.99% availability SLA, with multi-architecture (arm64/amd64) CI via Forgejo Actions
- Architected DNSSEC-signed, split-horizon DNS with catalog-based zone replication, and high-availability PostgreSQL (Patroni/TimescaleDB) with automated failover
- Instrumented full-stack observability (Grafana, Prometheus, VictoriaLogs, InfluxDB) and a Wazuh SIEM; root-caused and remediated live outages spanning DNS blackholes, container-runtime deadlocks, and memory-pressure cascades
- Contribute upstream to open-source infrastructure projects and publish GPG-signed Debian packages and npm modules through a self-hosted registry with multi-arch build pipelines

Founder & Principal Consultant

Brooklyn Helpdesk, Inc. — Brooklyn, NY

Jan 2018 - Present

- Deliver IT and DevOps engineering across ~500 managed client endpoints: IAM/SSO integration, CI/CD automation, and LAN/WiFi design, installation, and maintenance
- Standardized client identity on single sign-on (Okta, GSuite, Active Directory) with automated device lifecycle management via MDM, cutting onboarding and provisioning time ~30% on average across clients
- Deliver home and office automation with Home Assistant and Homebridge/HomeKit — presence-aware lighting, climate, window coverings, and A/V scene control over Zigbee, Z-Wave, and Matter

keepalived (VRRP) Backup & Recovery

SQL

LANGUAGES & SCRIPTING

Bash Python Shell Scripting Go

JavaScript TypeScript Node.js Vue

SQL HTML TOML / XML / YAML

PLATFORMS & MONITORING

Active Directory AWS Azure

Cloudflare Docker Elasticsearch

GCP GitHub Grafana InfluxDB

JAMF JumpCloud Kibana

Kubernetes Logstash Nagios Okta

Prometheus Splunk UniFi Network

UniFi Protect Zabbix Podman

Technitium DNS / DNSSEC Traefik

Pocket ID VictoriaLogs Wazuh (SIEM)

TimescaleDB

TECH STANDARDS & CONCEPTS

Agile Workflow CI/CD

Cloud Computing Directory Services

Distributed Computing DNS DNSSEC

Split-Horizon DNS High Availability

Reverse Proxy IAM LAMP Stack

Microservices Monitoring

Incident Response Linux Administration

MTTR On-Call Disaster Recovery (DR)

Backup & Restore Capacity Planning

Change Management Zero-Trust

Site-to-Site VPN Cost Optimization

Vendor Management

Remote Config & Management SaaS

SAML / OIDC Version Control

MCP (Model Context Protocol)

Director of IT & Infrastructure Engineer

Nov 2019 - Dec 2021

Symbiont.io — New York, NY

- Reduced onboarding time by 25% and costs by 12% via streamlined IAM/MDM integrations (Active Directory, GSuite, Okta, JAMF) to achieve SSO from hardware to cloud
- Designed and deployed automation software integrations for corporate IAM, user management, and hardware de/provisioning
- Deployed, monitored, and maintained Kubernetes and Docker stacks for a team of 36 engineers

Software Infrastructure Engineer & Interim Director of IT

Aug 2018 - Feb 2019

Blink Health — New York, NY

- Provisioned all Windows server instances with focus on Active Directory (fault tolerance, high availability) using Chef, Ansible, and Terraform
- Automated AD credential writing to GSuite, reducing onboarding steps from 40 to 4
- Leveraged Google SAML to reduce external attack surface and lower cost of SSO delivery
- Integrated Windows/Mac/Linux/GSuite auth via LDAP providing cross-platform SSO for VPNs, databases, and critical applications

Lead IT Infrastructure Engineer

Oct 2016 - Aug 2018

Blink Health — New York, NY

- Designed and managed UniFi 802.1X LAN/WiFi infrastructure at three sites with AWS connectivity via site-to-site VPN for 200+ staff
- Achieved infrastructure-as-code with GitHub PR support, CircleCI integration, and Terraform revision history across multiple AWS accounts
- Managed annual technology budget of \$5mm+ with inventory comprising hundreds of endpoints
- Grew IT team to 10 members across three geographic locations; oversaw opening of four offices in three states

AVP, Technology – Level III System & Network Engineer

Feb 2013 - May 2016

Build America Mutual Assurance Company — New York, NY

- Designed and implemented Active Directory backbone for SSO across Windows, Mac, and Linux using AD, Kerberos, LDAP, and MFA
- Implemented Jabber/XMPP with SSO and VOIP integration for the pre-existing phone system
- Migrated servers and applications to cloud providers, achieving \$250K+ in savings over five years
- Conducted Sarbanes-Oxley compliance reviews; supported nearly 100 users across three locations

Support Technician II

Jan 2012 - Jan 2013

SecondMarket, Inc. — New York, NY

- Configured SAML/ADFS SSO for Yammer, Salesforce, and Zendesk — saving the company \$36 per user per year
- Managed \$1.5M+ annual technology budget across three locations (New York, San Francisco, Hong Kong)